# Finite-Codimensional Invariance for Norming M-Bases

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## Source Problem

Hajek and Russo ask the following heredity problem for norming Markushevich bases in [1, Problem 6.1]: if a Banach space $X$ admits a norming M-basis and $Z$ is a subspace of $X$, must $Z$ admit a norming M-basis? The source problem appears on PDF page 24 of arXiv:2402.18442 and in the parsed source at `manuscript.tex:685-691`.

> distinct conditions in general. In other words, to the best of the authors' knowledge, the following problem is still open.
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> **Problem 6.1.** *Let $\mathcal{X}$ be a Banach space with norming M-basis and $\mathcal{Z}$ be a subspace of $\mathcal{X}$. Must $\mathcal{Z}$ admit a norming M-basis as well?*
>
> In this section we will discuss some particular cases where a positive answer is available

## Result

This packet proves that the answer is positive for all finite-codimensional subspaces. In fact, admitting a norming M-basis is invariant under passing to closed finite-codimensional subspaces.

**Theorem 1.** *Let $X$ be a Banach space with a norming M-basis. If $Z$ is a closed finite-codimensional subspace of $X$, then $Z$ admits a norming M-basis.*

**Theorem 2.** *Let $Z$ be a closed finite-codimensional subspace of a Banach space $X$. Then $X$ admits a norming M-basis if and only if $Z$ admits a norming M-basis.*

## Proof Intuition

There are two steps. First, if the finite-dimensional annihilator of $Z$ is already contained in the coordinate span of a given norming M-basis, Gaussian elimination removes finitely many pivot coordinates and leaves a norming M-basis on $Z$. Second, an arbitrary finite-codimensional subspace can be isomorphically moved to such a coordinate-annihilator subspace. Indeed, choose finitely many original basis vectors that span the finite-dimensional quotient $X/Z$. Because the coordinate span is norming, hence separates points, its restrictions to this finite-dimensional complement give all coordinate functionals on the complement. These coordinate-span functionals define a model subspace $W$ with a coordinate annihilator, and an explicit finite-rank formula gives an isomorphism $Z \cong W$.

# Proof

**Lemma 1.** *Let $\{e_\gamma;\varphi_\gamma\}_{\gamma\in\Gamma}$ be a $\lambda$-norming M-basis of a Banach space $X$, where $0<\lambda\le 1$. Let $E$ be a finite-dimensional subspace of $Y=\operatorname{span}\{\varphi_\gamma:\gamma\in\Gamma\}$, and set*

$$W=\bigcap_{f\in E}\ker f.$$

*Then $W$ admits a $\lambda$-norming M-basis.*

*Proof.* If $E=\{0\}$ there is nothing to prove, so assume $\dim E=m>0$. Choose a basis $f_1,\dots,f_m$ of $E$. Since $E$ is contained in the coordinate span, there is a finite set $A\subseteq\Gamma$ such that

$$E\subseteq\operatorname{span}\{\varphi_\alpha:\alpha\in A\}.$$

The $m\times|A|$ matrix $(f_i(e_\alpha))_{i,\alpha}$ has rank $m$. Choose $B\subseteq A$ with $|B|=m$ such that the square matrix $(f_i(e_\beta))_{i,\beta\in B}$ is invertible.

For each $\alpha\in\Gamma\setminus B$, let $(c_{\beta\alpha})_{\beta\in B}$ be the unique scalar family satisfying

$$f_i(e_\alpha)=\sum_{\beta\in B}c_{\beta\alpha}f_i(e_\beta)\qquad(i=1,\dots,m).$$

Define

$$u_\alpha=e_\alpha-\sum_{\beta\in B}c_{\beta\alpha}e_\beta\in W,\qquad\psi_\alpha=\varphi_\alpha|_W\quad(\alpha\in\Gamma\setminus B).$$

Then, for $\alpha,\eta\in\Gamma\setminus B$,

$$\psi_\alpha(u_\eta)=\varphi_\alpha(e_\eta)-\sum_{\beta\in B}c_{\beta\eta}\varphi_\alpha(e_\beta)=\delta_{\alpha\eta},$$

so the system is biorthogonal.

We next prove fundamentality. The above choice of $B$ defines a continuous finite-rank projection $P:X\to W$ by

$$Px=x-\sum_{\beta\in B}a_\beta(x)e_\beta,$$

where the scalars $a_\beta(x)$ are the unique solution of

$$f_i(x)=\sum_{\beta\in B}a_\beta(x)f_i(e_\beta)\qquad(i=1,\dots,m).$$

The functionals $a_\beta$ are continuous linear combinations of $f_1,\dots,f_m$. Hence $P$ is continuous, $PX=W$, and $Pe_\alpha=u_\alpha$ for $\alpha\notin B$. Since $\operatorname{span}\{e_\gamma:\gamma\in\Gamma\}$ is dense in $X$, it follows that $\operatorname{span}\{u_\alpha:\alpha\in\Gamma\setminus B\}$ is dense in $W$.

For totality, suppose $z\in W$ and $\psi_\alpha(z)=0$ for every $\alpha\in\Gamma\setminus B$. Then

$$z-\sum_{\beta\in B}\varphi_\beta(z)e_\beta$$

is annihilated by every $\varphi_\gamma$, $\gamma\in\Gamma$, and is therefore zero by totality of the original M-basis. Thus $z\in\operatorname{span}\{e_\beta:\beta\in B\}$. But $z\in W$ and the matrix $(f_i(e_\beta))_{i,\beta\in B}$ is invertible, so all coefficients are zero. Hence $z=0$.

It remains to check the norming constant. Let $Y_0=\operatorname{span}\{\psi_\alpha:\alpha\in\Gamma\setminus B\}\subseteq W^*$. The equations $f_i(z)=0$ for $z\in W$ and the invertibility of the pivot matrix express each omitted restriction $\varphi_\beta|_W$, $\beta\in B$, as a linear combination of the restrictions $\varphi_\alpha|_W$ with $\alpha\in A\setminus B$. Therefore $Y|_W=Y_0$.

Since $Y$ is $\lambda$-norming for $X$, for every $z \in W$,

$$\lambda \|z\|_X \le \sup\{|y^*(z)| : y^* \in Y,\ \|y^*\|_{X^*} \le 1\} \le \sup\{|g(z)| : g \in Y_0,\ \|g\|_{W^*} \le 1\}.$$

Thus $Y_0$ is $\lambda$-norming for $W$, and the biorthogonal system $\{u_\alpha; \psi_\alpha\}_{\alpha \in \Gamma \setminus B}$ is a $\lambda$-norming M-basis of $W$. $\square$

*Proof of Theorem 1.* Let $\{e_\gamma; \varphi_\gamma\}_{\gamma \in \Gamma}$ be a norming M-basis of $X$, and put $Y = \operatorname{span}\{\varphi_\gamma : \gamma \in \Gamma\}$. If $Z = X$, there is nothing to prove, so assume $m = \dim(X/Z) > 0$. Let $q : X \to X/Z$ be the quotient map. Since $\operatorname{span}\{e_\gamma : \gamma \in \Gamma\}$ is dense in $X$ and $X/Z$ is finite-dimensional, the vectors $q(e_\gamma)$ span $X/Z$. Choose $B = \{\beta_1, \dots, \beta_m\} \subseteq \Gamma$ such that $q(e_{\beta_1}), \dots, q(e_{\beta_m})$ is a basis of $X/Z$, and set $L = \operatorname{span}\{e_{\beta_1}, \dots, e_{\beta_m}\}$. Then $X = Z \oplus L$.

Let $f_1, \dots, f_m \in X^*$ be the coordinate functionals of this decomposition, so

$$f_i\Big(z + \sum_{j=1}^{m} a_j e_{\beta_j}\Big) = a_i \qquad (z \in Z).$$

The restrictions of $Y$ to $L$ separate points of $L$, because $Y$ separates points of $X$. Since $L$ is finite-dimensional, $Y|_L = L^*$. Hence there are $g_1, \dots, g_m \in Y$ such that

$$g_i(e_{\beta_j}) = \delta_{ij} \qquad (1 \le i, j \le m).$$

Define

$$W = \bigcap_{i=1}^{m} \ker g_i.$$

By Lemma 1, $W$ admits a norming M-basis.

It remains only to identify $Z$ and $W$ isomorphically. Define

$$S : Z \to W, \qquad Sz = z - \sum_{i=1}^{m} g_i(z) e_{\beta_i}.$$

The formula gives $g_j(Sz) = 0$ for all $j$, so $S$ maps $Z$ into $W$. Conversely, define

$$R : W \to Z, \qquad Rw = w - \sum_{i=1}^{m} f_i(w) e_{\beta_i}.$$

Then $R$ maps $W$ into $Z$. A direct computation using $g_i(e_{\beta_j}) = f_i(e_{\beta_j}) = \delta_{ij}$ gives $RS = I_Z$ and $SR = I_W$. Thus $Z$ is isomorphic to $W$. Pulling a norming M-basis of $W$ back through this isomorphism gives a norming M-basis of $Z$. $\square$

*Proof of Theorem 2.* The forward implication is Theorem 1. For the converse, assume that $Z$ admits a $\lambda$-norming M-basis $\{z_\alpha; z_\alpha^*\}_{\alpha \in A}$. Choose a finite-dimensional complement $L$ of $Z$ in $X$, so $X = Z \oplus L$, and let $P_Z : X \to Z$ and $P_L : X \to L$ be the associated bounded projections. Take a basis $u_1, \dots, u_n$ of $L$ with biorthogonal coordinate functionals $u_1^*, \dots, u_n^* \in L^*$.

Define functionals on $X$ by

$$\widetilde{z}_\alpha^* = z_\alpha^* \circ P_Z, \qquad \widetilde{u}_i^* = u_i^* \circ P_L.$$

Then

$$\{z_\alpha; \widetilde{z}_\alpha^*\}_{\alpha \in A} \cup \{u_i; \widetilde{u}_i^*\}_{i=1}^{n}$$

is a biorthogonal system in $X$. Its vectors are fundamental because $\operatorname{span}\{z_\alpha\}$ is dense in $Z$ and $u_1, \dots, u_n$ span $L$; its functionals are total because they separately detect the $Z$ and $L$ coordinates.

It remains to check that their linear span is norming. Let $D = \operatorname{span}\{z_\alpha^* : \alpha \in A\}$ and

$$\widetilde{D} = P_Z^* D + P_L^* L^* \subseteq X^*.$$

Since the norm $x = z + u \mapsto \|z\| + \|u\|$ is equivalent to the original norm on the finite direct sum $Z \oplus L$, and since $D$ is norming for $Z$ while $L^*$ is norming for $L$, there is a constant $c > 0$ such that

$$\sup\{|f(x)| : f \in \widetilde{D},\ \|f\|_{X^*} \le 1\} \ge c\|x\|_X \qquad (x \in X).$$

Thus $\widetilde{D}$ is a norming subspace of $X^*$, and the displayed biorthogonal system is a norming M-basis of $X$. □

## Limitations and Novelty Check

The theorem does not solve the full heredity problem: arbitrary infinite-codimensional subspaces are not covered. It also does not settle the complemented-subspace problem quoted later in the source paper.

The run indexes were checked for arXiv:2402.18442, norming Markushevich bases, the heredity problem, finite-codimensional subspaces, and hyperplanes. A bounded web search on 2026-06-15 used phrases combining norming M-bases or norming Markushevich bases with finite-codimensional subspaces, hyperplanes, subspaces, and the heredity problem. It did not find this exact subcase recorded as a later answer. After the finite-codimensional upgrade, the same search phrases were rerun with “finite codimension”, “finite-codimensional”, “hyperplane”, “separable quotient”, and “quotient” variants; no exact duplicate was found.

## References

## References

[1] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.